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Prevalence, associated factors, and machine learning-based prediction of depression, anxiety, and stress among university students: a cross-sectional study from Bangladesh

Md Emran Hasan[1,2], Mohammad Arif[1,3], S. M. Rakibul Hasan[1,4,11], Moses Muwanguzi[5], Joan Abaatyo[6,7], Mark Mohan Kaggwa[8], Moneerah Mohammad ALmerab[9], Pawel A. Atroszko[10], Mohammad Muhit[11], Firoj Al-Mamun[1,11,12] and Mohammed A. Mamun[1,11,12*]

Abstract

Background Mental health challenges are a growing global public health concern, with university students at elevated risk due to academic and social pressures. Although several studies have exmanined mental health among Bangladeshi students, few have integrated conventional statistical analyses with advanced machine learning (ML) approaches. This study aimed to assess the prevalence and factors associated with depression, anxiety, and stress among Bangladeshi university students, and to evaluate the predictive performance of multiple ML models for those outcomes.

Methods A cross-sectional survey was conducted in February 2024 among 1697 students residing in halls at two public universities in Bangladesh: Jahangirnagar University and Patuakhali Science and Technology University. Data on sociodemographic, health, and behavioral factors were collected via structured questionnaires. Mental health outcomes were measured using the validated Bangla version of the Depression, Anxiety, and Stress Scale-21 (DASS-21). Statistical analyses included chi-square tests and binary logistic regression, while seven ML models including, K-Nearest Neighbors (KNN), Random Forest (RF), Gradient Boosting Machine (GBM), Extreme Gradient Boosting (XGBoost), Categorical Boosting (CatBoost), Logistic Regression (LR), and Support Vector Machine (SVM) were employed to predict mental health outcomes.

Results The prevalence of depression, anxiety, and stress was 56.9%, 69.5%, and 32.2%, respectively. Significant associated factors for depression included unfriendly family relationships, enrollment in commerce, and cigarette smoking. Female gender, unfriendly family relationships, academic year, and cigarette smoking were significant factors for stress. No significant factors were identified for anxiety. Among ML models, SVM achieved the highest accuracy for depression prediction (accuracy = 0.5693; precision = 0.7560; log loss = 0.6847), LR for anxiety (accuracy = 0.6948; precision = 0.7881), and CatBoost for stress (accuracy = 0.6706; precision = 0.6454; F1-score = 0.5777; log loss = 0.6284).

*Correspondence:
Mohammed A. Mamun
mamun@thechinta.org

Full list of author information is available at the end of the article

Feature importance analyses highlighted faculty of study and relation with family as the top predictors. ROC-AUC values indicated moderate discriminatory performance (all ≥ 0.5).

Conclusions Integrating machine learning with conventional analyses enhances the identification and prediction of factors associated with depression, anxiety, and stress among university students. These findings support the implementation of campus-based mental health screening, accessible counseling, and peer support programs, and highlight the value of data-driven approaches for developing targeted university mental health policies.

Keywords Mental health, Machine learning, University students, Gradient boosting machines, SHAP

Introduction

Mental health is a crucial component of overall wellbeing, encompassing an individual's psychological, emotional, and social state. According to the World Health Organization [1], mental health is defined as a state in which individuals can recognize their potential, manage daily stressors, function productively, and contribute to their community. Despite its importance, mental health conditions are increasingly prevalent worldwide, particularly among young adults such as university students [2]. These conditions pose significant public health challenges, often resulting in impaired academic performance, social isolation, and heightened suicide risks [1, 3].

University students are particularly vulnerable to mental health issues due to academic pressures, financial stress, and social transitions. A study in Pakistan found anxiety to be highly prevalent (88.4%), followed by depression (75%) and stress (84.4%) among university students [4]. Similarly, a review in the United States reported that 22% of undergraduate students experienced depression, with 8% suffering from PTSD and 2–12% affected by compulsive disorders [5]. Alarmingly, severe or extremely severe levels of depression, anxiety, and stress have been reported among both students and university staff, with prevalence rates exceeding 30% in certain groups [6]. Students in health sciences have shown even higher rates, with 65% experiencing stress, 85.1% reporting anxiety, and 51.4% facing depression [7]. In low- and middle-income countries (LMICs), where nearly half of the world's youth population resides, mental health concerns are particularly pronounced. A systematic review and meta-analysis found that approximately 24.4% of university students in LMICs reported symptoms of depression [2]. Similar findings were observed in Chinese university students, with a reported depression prevalence of 23.8% [8].

Numerous factors contribute to the rising burden of mental health conditions among university students. A cross-sectional study in Malaysia identified key predictors of anxiety, including academic year, financial instability, alcohol consumption, poor sleep quality, body mass index, and strained relationships with peers and lecturers [9]. Similarly, mental health conditions have been linked to poor academic performance, highlighting the broader implications of these issues on students' educational outcomes [3]. Sociodemographic and behavioral factors such as age, insomnia, substance use, and social isolation further compound mental health risks, as demonstrated in studies among university students in Spain and other regions [10]. Substance use further compounds the risk of mental health issues. For instance, a study among Canadian Indigenous students revealed that alcohol, marijuana, and recreational drug use significantly increased vulnerability to depression, anxiety, and suicidal behaviors [11].

Despite growing evidence on mental health issues among university students, studies specifically focusing on depression, anxiety, and stress status among Bangladeshi students, particularly those identifying key risk factors using predictive models, remain limited. Existing research provides some insights into these associations. For instance, one study found that Bangladeshi university students reported mental health problems with a prevalence rate from 25% to 71%. Students from extended families had a lower likelihood of experiencing depression and stress, with respective relative risk ratios of 0.68 and 0.65. Conversely, students from business faculties had a 2.28 times higher rate of stress compared to their peers in science and engineering disciplines [12]. Additionally, 40% of students were found to experience extremely severe anxiety, with gender and family type playing notable roles in this outcome [13]. Several behavioral and social factors have also been linked to mental health problems in the country. Studies have identified internet addiction [14, 15], loneliness, physical inactivity [16], problematic internet use [17, 18], and cyber victimization [19] as significant factors of psychological distress. Notably, smoking has been strongly associated with deteriorated mood and anxiety symptoms, with 70.3% of smokers reporting negative mental health impacts [20]. Although limited, some studies have explored the association between substance use and mental health in Bangladesh, revealing that depression is a key factor of psychoactive substance dependence among students [21].

Despite these insights, research employing machine learning (ML) techniques to predict mental health risks in Bangladeshi students is still in its early stages. One study employing the XGBoost model identified social media addiction, age, academic performance, smoking status, monthly family income, and morningness-eveningness as primary risk factors for anxiety, depression, and insomnia [22]. Another study reported stress among nearly one-third of students, with Random Forest models performing best in predicting risk factors such as students' pulse rate, systolic and diastolic blood pressure, sleep patterns, smoking habits, and academic background [23]. Moreover, a hybrid stacking model achieved impressive predictive accuracy rates of 99% for depression and 97% for anxiety among Bangladeshi university students [24]. In addition to these findings, an association rule mining study using the Apriori algorithm identified various psychosocial factors, such as academic pressure (72.4%), uncertainty about the future (56.3%), hopelessness (48.3%), family expectations (47.1%), financial crisis (42.5%), loneliness (41.4%), and unemployment (37.9%), as strong predictors of depression and suicidal ideation among Bangladeshi students [25]. However, while one study employed ML models to classify depression, anxiety, and stress with an impressive AUROC score of 0.98, 0.97, and 0.97, respectively, it lacked an in-depth exploration of associated factors, particularly during the COVID-19 pandemic [26].

While a growing body of research has explored mental health problems among Bangladeshi university students—including during the COVID-19 period—most studies have relied on conventional statistical techniques, smaller or single-institution samples, and limited scope [27]. As a result, there remains a lack of large-scale, comprehensive studies that integrate both traditional epidemiological analyses and advanced machine learning methods to investigate multiple mental health outcomes and their predictors within this population. This study aims to address this gap by investigating the prevalence and associated factors of these mental health conditions while employing ML models such as CatBoost, XGBoost, SVM, RF, KNN, and GBM for enhanced predictive accuracy. By integrating these approaches, this study seeks to provide deeper insights into complex risk patterns, offering robust evidence to guide targeted interventions. The findings will inform policymakers, university authorities, and mental health professionals in developing effective strategies to improve student well-being in Bangladesh and other low- and middle-income countries.

Methods

Study participants and procedure

This cross-sectional study targeted university students in Bangladesh and collected data from two public universities, specifically from Jahangirnagar University, Dhaka, and Patuakhali Science and Technology University, Patuakhali. Data collection was conducted in university residential halls by two distinct teams, each with 12 trained research assistants, in February 2024. Eligible participants were currently enrolled undergraduate or graduate students aged 18 years or older; whereas students on leave, visiting students, and internship-only enrollees were excluded. In the absence of a complete sampling frame, convenience sampling with predefined stratified recruitment approach by faculty (Arts, Science, Commerce), academic year, and gender was implemented to enhance heterogeneity. Data were collected in person using structured, self-administered, paper-based questionnaires, distributed in residence halls and classrooms with the cooperation of university staff. Before participation, all students were briefed about the study's objectives, procedures, and their rights, and written informed consent was obtained. The survey typically took 15–20 min to complete. After collection, all responses were manually entered into a secure Google Forms database by trained research assistants. To ensure data quality, each entry was independently cross-checked by another team member. Incomplete, duplicate, or inconsistent responses were excluded from the final dataset. Specifically, any response missing at least one outcome variable was excluded from analysis. After applying these quality control measures, a total of 1,697 complete and valid responses were retained for final analysis. Initially, data were collected from 1,709 students, both from classrooms and university residential halls. All available and willing students present during the data collection period were approached in person by trained research assistants. However, due to institutional constraints, the exact number of students residing in the halls was not available; as such, the formal response rate could not be calculated.

Measures

Sociodemographic factors

Participants provided socio-demographic information, including gender, age, marital status, socioeconomic status, family type, and current living location. Family type was classified as either a nuclear family, defined as a household comprising parents and their children, or a joint family, characterized by the cohabitation of multiple generations and extended family members within a single household. Socioeconomic status determined using monthly family income was categorized into lower class (earning less than 15,000 Bangladeshi Taka [BDT]), middle class (earning from 15,000 to 30,000 BDT), and upper class (earning more than 30,000 BDT). Relationships with the family were categorized as friendly or not.

Health and behavioral variables

The study examined instances of cigarette smoking, illicit drug use, and alcohol consumption to assess different types of risk-related behaviors. Participants reported their history with each behavior, indicating whether they had ever engaged in cigarette smoking, illicit drug use, or alcohol consumption [28, 29]. Responses were dichotomous (yes/no).

Mental health problems

The Depression Anxiety Stress Scale (DASS) is a 21-item self-report inventory that assesses three subscales: depression (7 items), anxiety (7 items), and stress (7 items) [30]. Each item is rated on a four-point Likert scale ranging from "never" (0) to "always" (3). In this study, the Bangla version of the DASS-21 was utilized [31]. For each subscale, the scores of the seven relevant items were summed and then multiplied by two to obtain the final score, following standard DASS-21 guidelines. The cut-off scores for the validated Bangla DASS-21 severity bands were used as follows: depression, 0–9 normal, 10–13 mild, 14–20 moderate, 21–27 severe, ≥28 extremely severe; anxiety, 0–7 normal, 8–9 mild, 10–14 moderate, 15–19 severe, ≥20 extremely severe; stress, 0–14 normal, 15–18 mild, 19–25 moderate, 26–33 severe, ≥34 extremely severe (149–154). For binary classification in statistical analyses, participants were defined as 'cases' when their subscale score was moderate or higher (depression ≥ 14; anxiety ≥ 10; stress ≥ 19); otherwise, they were classified as 'non-cases' [30, 31]. In the Bangla version, the Cronbach's alpha coefficients for the depression, anxiety, and stress subscales were found to be 0.99, 0.96, and 0.96, respectively [31]. In the present study, the Cronbach's alpha coefficients for these subscales were 0.787, 0.783, and 0.834, respectively, with an overall the Cronbach's alphaof 0.904.

Ethical consideration

The study meticulously followed ethical guidelines outlined in the Helsinki Declaration of 2013 to safeguard the welfare of human participants. Ethical clearance was obtained from the review board at Patuakhali Science and Technology University [Reference: PSTU/ICE/2023/83], before the study's initiation. Before recruitment, all participants were provided with comprehensive written and verbal information outlining the study's objectives, procedures, potential risks and benefits, as well as assurances regarding confidentiality and anonymity. Participants were informed that their involvement was entirely voluntary, and they could decline or withdraw from the study at any point without any consequences. Written informed consent was obtained from each participant before data collection. All data were collected and stored confidentially. Identifiable personal information was not recorded, and responses were anonymized at the point of data entry. Only authorized research team members had access to the dataset. These procedures were implemented to ensure privacy and uphold the highest standards of research ethics throughout the study. No monetary or non-monetary incentives were provided.

Statistical analysis

For this study, data were entered into Google Forms, followed by formatting for an SPSS file format to facilitate final analysis. Descriptive and inferential statistics were employed to analyze the data. Descriptive statistics such as frequency and percentage calculations were used to summarize participant characteristics and the prevalence of depression, anxiety, and stress. Inferential ststistics included both bivariate and multivariate analyses. To examine associations between potential factors and binary mental health outcomes (depression, anxiety, and stress status), we first conducted bivariate analyses using the chi-square test. The chi-square test was selected because all outcome variables were binary, and the chi-square test is appropriate for assessing associations between categorical or ordinal predictors and binary outcomes in this context.

For multivariable analysis, binary logistic regression models were constructed for each mental health outcome (depression, anxiety, and stress). All models were adjusted for related covariates, such as gender, marital status, family type, relation with family, cigarette smoking, illicit drug use, and alcohol use. Results are presented as adjusted odds ratios (AORs) with 95% confidence intervals. To ensure the validity of the logistic regression models, we assessed multicollinearity among the predictor variables by calculating variance inflation factors (VIFs). All VIF values were found to be within acceptable limits (< 5), indicating no evidence of problematic multicollinearity. A significance level of $p < 0.05$ was set for all tests, and a standard 95% confidence interval was utilized throughout the study. All statistical analyses were conducted using SPSS version 27 (IBM Corp., Armonk, NY, USA).

Machine learning analysis

In this work, depression, anxiety, and stress, with their associated factors among university students, were analyzed and predicted using machine learning techniques. Python, the primary programming language, was used in conjunction with Google Colab to analyze the data. The dataset was divided into two categories: 80% for training and 20% for testing. To address class imbalance in the training data, Synthetic Minority Over-sampling Technique (SMOTE) was applied exclusively to the training dataset. SMOTE was utilized to generate synthetic examples of the minority class, ensuring that the class distribution was more balanced and that the models could learn effectively from both classes. For this research, the prediction capabilities of various machine learning models were evaluated, specifically K-Nearest Neighbors

(KNN), Random Forest (RF), Extreme Gradient Boosting (XGBoost), Categorical Boosting (CatBoost), Logistic Regression (LR), Gradient Boosting Machines (GBM), and Support Vector Machine (SVM). These models were selected to provide a comprehensive comparison of different algorithmic approaches in addressing the classification task at hand. All models were trained using 5-fold cross-validation to ensure robust evaluation and reduce the risk of overfitting. The average performance across all five folds was reported to obtain the final results. To optimize the model performance, hyperparameter tuning was performed for all models. Grid search and random search were used to find the best combination of hyperparameters for each algorithm. To assess the importance of features for outcome, feature extraction techniques were employed. While XGBoost assessed feature importance, improving model interpretability and performance, CatBoost used SHAP values to explain the impact of key features on model predictions, providing a clearer understanding of the factors influencing outcomes. In addition to accuracy, precision, F1-score, and log-loss metrics, each model's performance was assessed using the area under the receiver operating characteristic curve (AUC-ROC).

Classification tasks in health-related datasets are well-suited for machine learning models, particularly tree-based and boosting algorithms such as RF, XGBoost, CatBoost, and GBM. These models can effectively detect complex, nonlinear relationships and interactions between features, an advantage over traditional statistical models that might not capture such dynamics. In the context of this study, these tree-based models were chosen due to their ability to handle high-dimensional data and complex relationships inherent in mental health outcomes such as depression, anxiety, and stress. Additionally, these models are known for their robustness in scenarios with relatively balanced class distributions, as seen in our dataset, which has a moderate sample size. Hyperparameter tuning was applied to all models, including XGBoost and SVM, using grid search and random search techniques to identify the best set of parameters for each algorithm. This tuning process was critical for improving model performance and ensuring the comparison was done under optimal conditions. Furthermore, feature extraction techniques such as SHAP for CatBoost and Gini importance for XGBoost were utilized to gain insights into the importance of specific features in predicting mental health outcomes. All models were evaluated based on five-fold cross-validation, ensuring the reliability and generalizability of the results.

Machine learning models

K-Nearest neighbors (KNN)

KNN is an instance-based learning technique known for its deferred computation until after classification. Operating as a non-parametric approach, it employs locally approximated functions for regression and classification tasks [32]. KNN identifies the K training examples most like the object to be classified in the feature space. The object's class membership is determined by a majority vote among its K nearest neighbors, typically using a modest positive integer value for K. When K equals 1, the object is assigned to the class of its single nearest neighbor [33].

Random forest (RF)

The RF algorithm is a robust ensemble learning technique designed for both classification and regression tasks. It generates a multitude of decision trees during the training phase [34]. Unlike traditional decision tree methods, RF incorporates bagging and random feature selection to build a forest of trees, collectively producing superior forecasts compared to any single tree. By averting overfitting, a common issue with individual decision trees, the ensemble approach significantly enhances model accuracy. Random Forest has proven effective and adaptable across various domains, demonstrating its efficacy in managing complex datasets comprising categorical and numerical variables [34].

Extreme gradient boosting (XGBoost)

XGBoost surpasses traditional gradient-boosting machines (GBMs) with its innovative gradient-boosting architecture, representing a significant advancement in ensemble machine learning techniques. This approach facilitates sequential decision tree construction, systematically reducing errors by learning from preceding tree mistakes [35]. XGBoost's predictive accuracy outperforms its GBM predecessors due to its systematic improvements in speed, performance, and adept handling of large-scale datasets [36].

Categorical boosting (CatBoost)

CatBoost is an advanced machine learning method designed to handle categorical data efficiently. Focused on decision tree gradient boosting, CatBoost addresses common issues associated with categorical data without extensive preprocessing. By combining one-hot encoding with innovative algorithmic techniques, CatBoost improves prediction accuracy while mitigating overfitting. Its efficiency and scalability make CatBoost valuable across various applications, including recommendation systems and predictive modeling, particularly in datasets with prominent categorical features [37].

Gradient boosting machines (GBM)

GBM construct robust learners by sequentially adding weak learners, typically decision trees, resulting in improved predictive accuracy. This technique employs gradient descent to minimize the loss function, iteratively focusing on areas where previous models underperformed. GBMs are versatile and effective across diverse predictive applications, requiring meticulous hyperparameter tuning to manage overfitting and computational resources. Their ability to handle complex, nonlinear data makes GBMs widely applicable across various industries [36].

Support vector machine (SVM)

SVM is a primary supervised learning technique extensively used in regression and classification tasks. It segregates different class labels by identifying the optimal hyperplane in a high-dimensional space. Support vectors, critical data points closest to the hyperplane, significantly enhance SVM's predictive capacity. Kernel functions enable SVM to handle nonlinear data connections efficiently, contributing to its accuracy and versatility across complex datasets and multiple domains [38].

Logistic regression (LR)

LR is a widely used statistical method for binary classification, where it models the probability of a binary outcome based on a linear combination of input features. By applying the logistic (sigmoid) function, LR maps the predicted values to a probability between 0 and 1, which can then be classified into two categories based on a predefined threshold, usually 0.5. The model is trained by optimizing the parameters (weights) using maximum likelihood estimation, ensuring the best fit to the data [39]. In this study, LR was employed to predict depression, anxiety, and stress statuses, leveraging its simplicity, interpretability, and ability to provide insights into the influence of various predictors. LR is particularly effective in scenarios like this where the goal is to understand the relationship between mental health conditions and factors such as demographics and behavioral patterns. Its performance and interpretability make it an ideal choice for predicting mental health outcomes, as it allows researchers to estimate odds ratios, indicating the strength of the association between predictors and the outcomes [40].

Results

Description of the study participants

Table 1 shows the sociodemographic variables and their association with mental health status. Of the 1697 respondents, 59.9% were male and 40.1% were female, respectively. More than one-third (40.4%) of the study population were aged 21 to 22 years, with about 93.3% being unmarried. Different levels of the socio-economic class were observed as low (32.2%), middle (41.7%), and high (23.9%). Among participants, almost three-quarters (72.4%) belonged to a nuclear family background, and a majority of participants (89.8%) described their relationship with the family as friendly. More than two-thirds of participants (69.3%) were in their first year of study, with more than half (58.2%) studying under the faculty of science, and about a fifth (22.5%) of the participants reported cigarette smoking.

Prevalecne and severity of depression, anxiety, and stress

Figure 1 illustrates the distribution of mental health symptoms among the 1,697 university students surveyed. The prevalence of depression, anxiety, and stress was 56.9% ($n = 966$), 69.5% ($n = 1{,}179$), and 33.2% ($n = 564$), respectively. Regarding severity, 28.6% of students reported normal levels of depression, while 14.4% had mild, 29.4% moderate, 16.6% severe, and 11.0% extremely severe depressive symptoms. For anxiety, 24.0% of students were classified as normal, 6.5% as mild, 25.4% as moderate, 15.8% as severe, and a notably high 28.3% as extremely severe. In terms of stress, 50.4% of students reported normal levels, 16.4% had mild, 18.3% moderate, 10.9% severe, and 4.0% extremely severe symptoms.

Associations of sociodemographic and health-related factors with mental health outcomes

Gender was significantly associated with both depression ($\chi^2 = 3.863$, $p = 0.049$) and stress ($\chi^2 = 6.727$, $p = 0.009$). Interestingly, a higher proportion of males experienced depression compared to females (61.9% vs. 38.1%), while females had higher proportion of stress (55.5% vs. 44.5%). Regarding marital status, a significant association was found with anxiety ($\chi^2 = 5.856$, $p = 0.016$), with unmarried individuals showing a significantly higher prevalence of anxiety. However, no significant associations were found with depression ($\chi^2 = 0.597$, $p = 0.440$) and stress ($\chi^2 = 0.314$, $p = 0.575$). In terms of familial relationships, a significant association was observed across all mental health outcomes, depression ($\chi^2 = 15.623$, $p < 0.001$), anxiety ($\chi^2 = 5.897$, $p = 0.015$), and stress ($\chi^2 = 28.110$, $p < 0.001$). Depression and stress were significantly more prevalent among those reporting unfriendly familial relationships (both $p < 0.001$), while anxiety was also significantly more common in this group ($p = 0.015$). A significant association was found between studying year and stress, with those in their first year showing significantly higher rates of the outcome compared to students in later years ($\chi^2 = 11.179$, $p = 0.011$). Additionally, the faculty of study was also significantly associated with both depression ($\chi^2 = 28.669$, $p < 0.001$) and anxiety ($\chi^2 = 7.302$, $p < 0.026$), as students from the Science faculty showed significantly different rates of the outcome compared to those in Arts

Table 1 Characteristics of the participants and distributions of mental health status

Variables	Total Sample	Depression ($n=966$; 56.9%)		Anxiety ($n=1179$; 69.5%)		Stress ($n=564$; 33.2%)	
	n; (%)	Yes; *n* (%)	χ^2 (*p*-value)	Yes; *n* (%)	χ^2 (*p*-value)	Yes; *n* (%)	χ^2 (*p*-value)
Socio-demographic variables							
Age group							
18–20 years	613, 36.1%	335, 37.1%	2.893 (0.235)	421, 38.1%	0.414 (0.813)	193, 36.5%	1.683 (0.431)
21–22 years	686, 40.4%	389, 43.0%		473, 42.8%		239, 45.2%	
> 22 years	297, 17.5%	180, 19.9%		210, 19.0%		97, 18.3%	
Gender							
Male	1016, 59.9%	598, 61.9%	3.863 (**0.049**)	715, 60.6%	0.964 (0.326)	313, 55.5%	6.727 (**0.009**)
Female	681, 40.1%	368, 38.1%		464, 39.4%		251, 44.5%	
Marital status							
Unmarried	1584, 93.3%	895, 94.3%	0.597 (0.440)	1089, 93.8%	5.856 (**0.016**)	526, 95.1%	0.314 (0.575)
Married	89, 5.2%	54, 5.7%		72, 6.2%		27, 4.9%	
Socio-economic status							
Low	547, 32.2%	321, 33.8%	1.083 (0.582)	381, 33.0%	0.155 (0.926)	171, 31.1%	3.107 (0.212)
Middle	707, 41.7%	405, 42.6%		495, 42.9%		230, 41.9%	
High	405, 23.9%	224, 23.6%		279, 24.2%		148, 27.0%	
Family type							
Nuclear	1228, 72.4%	696, 73.3%	0.195 (0.659)	854, 73.8%	0.007 (0.935)	408, 74.2%	0.078 (0.780)
Joint	437, 25.8%	253, 26.7%		303, 26.2%		142, 25.8%	
Currently living location							
University hall	1451, 85.5%	815, 85.6%	4.349 (0.114)	1009, 86.5%	1.447 (0.485)	467, 84.0%	4.571 (0.102)
Outside varsity	102, 6.0%	68, 7.1%		75, 6.4%		40, 7.2%	
Others*	124, 7.3%	69, 7.2%		82, 7.0%		49, 8.8%	
Relationship with family							
Friendly	1524, 89.8%	847, 89.3%	15.623 (**< 0.001**)	1047, 90.5%	5.897 (**0.015**)	480, 86.5%	28.110 (**< 0.001**)
Unfriendly	140, 8.2%	102, 10.7%		110, 9.5%		75, 13.5%	
Studying year							
First	1176, 69.3%	674, 70.6%	1.435 (0.697)	818, 70.4%	2.029 (0.566)	380, 68.2%	11.179 (**0.011**)
Second	273, 16.1%	153, 16.0%		185, 15.9%		98, 17.6%	
Third	162, 9.5%	96, 10.1%		119, 10.2%		67, 12.0%	
Fourth	61, 3.6%	31, 3.2%		40, 3.4%		12, 2.2%	
Faculty of study							
Science	988, 58.2%	558, 61.7%	28.669 (**< 0.001**)	682, 61.6%	7.302 (**0.026**)	322, 60.6%	1.645 (0.439)
Arts	450, 26.5%	224, 24.8%		298, 26.9%		147, 27.7%	
Commerce	165, 9.7%	122, 13.5%		128, 11.6%		62, 11.7%	
Health and behavioral variables							
Cigarette smoking							
No	1315, 77.5%	718, 74.3%	12.859 (**< 0.001**)	898, 76.2%	3.878 (**0.049**)	416, 73.8%	6.741 (**0.009**)
Yes	382, 22.5%	248, 25.7%		281, 23.8%		148, 26.2%	
Illicit drug use							
No	1542, 90.9%	857, 88.7%	12.489 (**< 0.001**)	1057, 89.7%	6.859 (**0.009**)	502, 89.0%	3.518 (0.061)
Yes	155, 9.1%	109, 11.3%		122, 10.3%		62, 11.0%	
Alcohol use							
No	1536, 90.5%	861, 89.1%	4.989 (**0.026**)	1057, 89.7%	3.330 (0.068)	503, 89.2%	1.736 (0.188)
Yes	161, 9.5%	105, 10.9%		122, 10.3%		61, 10.8%	

Footnote: Others*: Students alternate between university residential hall and off-campus living

and Commerce faculties. In terms of health and behavioral factors, cigarette smoking was significantly associated with depression (χ^2 = 12.859, $p<0.001$), anxiety (χ^2 = 3.878, $p=0.049$), and stress (χ^2 = 6.741, $p=0.009$), with non-smokers exhibiting consistently higher prevalence rates. Similarly, illicit drug use was significantly associated with depression (χ^2 = 12.489, $p<0.001$) and anxiety (χ^2 = 6.859, $p=0.009$), while its association with stress was not significant (χ^2 = 3.518, $p=0.061$). Furthermore, alcohol use showed a significant association with depression (χ^2 = 4.989, $p=0.026$) and anxiety ($p=0.068$),

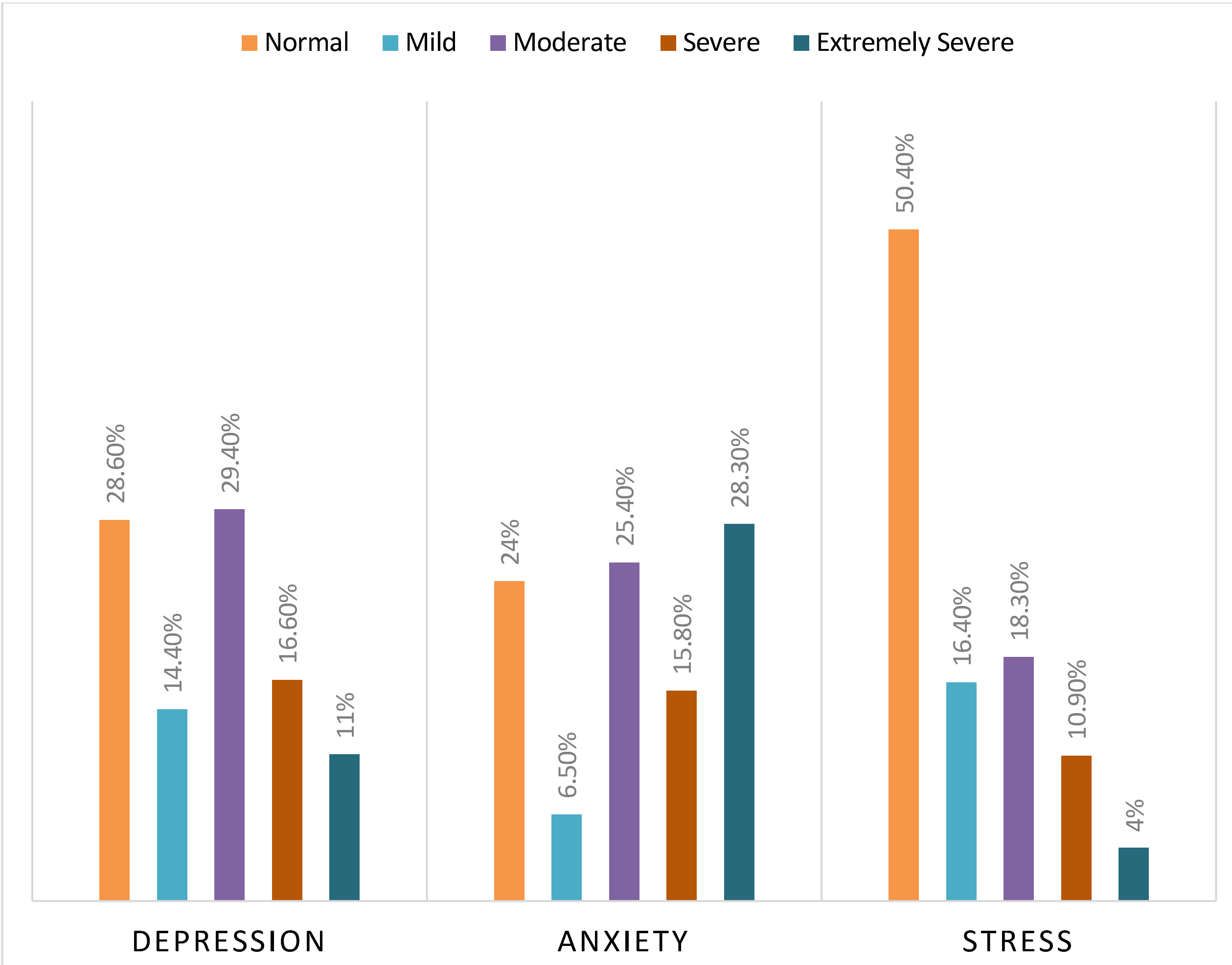

Fig. 1 Severity distribution of depression, anxiety, and stress among university students ($N = 1{,}697$)

while its associations with stress ($p = 0.188$) were not statistically significant (Table 1).

Factors associated with mental health outcomes

The adjusted logistic regression models revealed several significant factors of mental health outcomes among the participants (Table 2). Gender, specifically being female, was associated with a 1.588-fold higher odds of stress ($p = 0.001$), while family relationships showed that unfriendly relationships increased the likelihood of both depression (1.615 -fold, $p = 0.031$) and stress (1.943-fold, $p = 0.002$). Compared to first-year students, second, third, and fourth-year students showed varying levels of risk for stress, with odds ratios of 1.136, 1.074, and 0.344, respectively ($p = 0.039$). Students in the arts and commerce faculties were more likely to experience depression than science students ($p < 0.0001$ for both). Finally, cigarette smoking was identified as a significant factor for both stress (1.652-fold, $p = 0.003$) and depression (1.450 -fold, $p = 0.026$), but illicit drug use and alcohol use did not show significant associations with the mental health outcomes in this analysis.

Feature selection

The SHAP (Shapley Additive exPlanations) values for each feature are shown in Fig. 2, which shows how each feature contributes to the model's predictions. The SHAP value is shown by the horizontal axis; negative values cause the model output to decrease, while positive ones cause it to rise. Each dot, which is color-coded according to the feature's value (from low to high), represents a single observation. Features like faculty of study, cigarette smoking status, and relationship with family have a significant effect on the model; greater SHAP values indicate a bigger influence. On the other hand, characteristics such as marital status, alcohol usage status and currently living with are less significant. As indicated by CatBoost SHAP values, this graphic emphasizes the significance of particular parameters in the predictive model and shows

Table 2 Adjusted logistic regression models predicting mental health outcomes

Variables	Depression		Anxiety		Stress	
	AOR (95% CI)	*p*-value	AOR (95% CI)	*p*-value	AOR (95% CI)	*p*-value
Socio-demographic variables						
Age group						
18–20 years	1	0.343	1	0.734	1	0.693
21–22 years	1.055 (0.822–1.355)		1.045 (0.802–1.361)		1.073 (0.823-1.400)	
> 22 years	1.350 (0.900-2.027)		1.188 (0.773–1.826)		1.589 (0.786–1.824)	
Gender						
Male	1	0.349	1	0.326	1	**0.001**
Female	1.124 (0.880–1.438)		1.139 (0.878–1.479)		1.588 (1.223–2.063)	
Marital status						
Unmarried	1	0.123	1	0.045	1	0.510
Married	1.532 (0.891–2.636)		1.914 (1.015–3.610)		0.828 (0.473–1.450)	
Socio-economic status						
Low	1	0.718	1	0.826	1	0.772
Middle	1.001 (0.773–1.296)		1.089 (0.829–1.430)		1.094 (0.830–1.443)	
High	0.900 (0.669–1.212)		1.061 (0.775–1.454)		1.104 (0.807–1.511)	
Family type						
Nuclear	1	0.323	1	0.525	1	0.215
Joint	0.879 (0.681–1.135)		0.916 (0.700–1.200)		0.841 (0.639–1.106)	
Currently living location						
University hall	1	0.157	1	0.449	1	0.108
Outside varsity	1.525 (0.941–2.472)		1.244 (0.742–2.085)		1.419 (0.886–2.273)	
Others	0.861 (0.569–1.304)		0.826 (0.536–1.273)		1.441 (0.941–2.208)	
Relationship with family						
Friendly	1	**0.031**	1	0.480	1	**0.002**
Unfriendly	1.615 (1.045–2.497)		1.178 (0.748–1.855)		1.943 (1.287–2.933)	
Studying year						
First	1	0.084	1	0.591	1	**0.039**
Second	0.950 (0.688–1.311)		0.884 (0.629–1.242)		1.136 (0.815–1.583)	
Third	0.634 (0.396–1.014)		0.850 (0.515–1.402)		1.074 (0.667–1.729)	
Fourth	0.482 (0.253–0.921)		0.644 (0.329–1.261)		0.344 (0.151–0.782)	
Faculty of study						
Science	1	**< 0.001**	1	0.133	1	0.246
Arts	0.695 (0.544–0.890)		0.900 (0.694–1.169)		0.907 (0.696–1.182)	
Commerce	2.164 (1.435–3.265)		1.436 (0.940–2.193)		1.302 (0.885–1.917	
Health and behavioral variables						
Cigarette smoking						
No	1	**0.026**	1	0.341	1	**0.003**
Yes	1.450 (1.045–2.010)		1.182 (0.837–1.669)		1.652 (1.187–2.299)	
Illicit drug use						
No	1	0.709	1	0.502	1	0.796
Yes	1.127 (0.601–2.114)		1.258 (0.643–2.461)		0.922 (0.499–1.703)	
Alcohol use						
No	1	0.939	1	0.989	1	0.396
Yes	1.023 (0.571–1.833)		0.995 (0.537–1.847)		1.284 (0.721–2.286)	

how social and personal habits have distinct effects on the result.

The feature importance values produced by the XGBoost model are shown in Fig. 3, which shows how much each feature contributes to the model's output. The gain, or F score, that represents each feature's contribution to the prediction model is displayed on the horizontal axis. Relationships with family and gender are the most important features, with gain scores of 1.25 and 0.85, respectively, indicating that they have a significant influence on the model's predictions. Faculty of study, studying year, and socio-economic status are other noteworthy features that all significantly contribute to gaining value. On the other hand, marital status and family

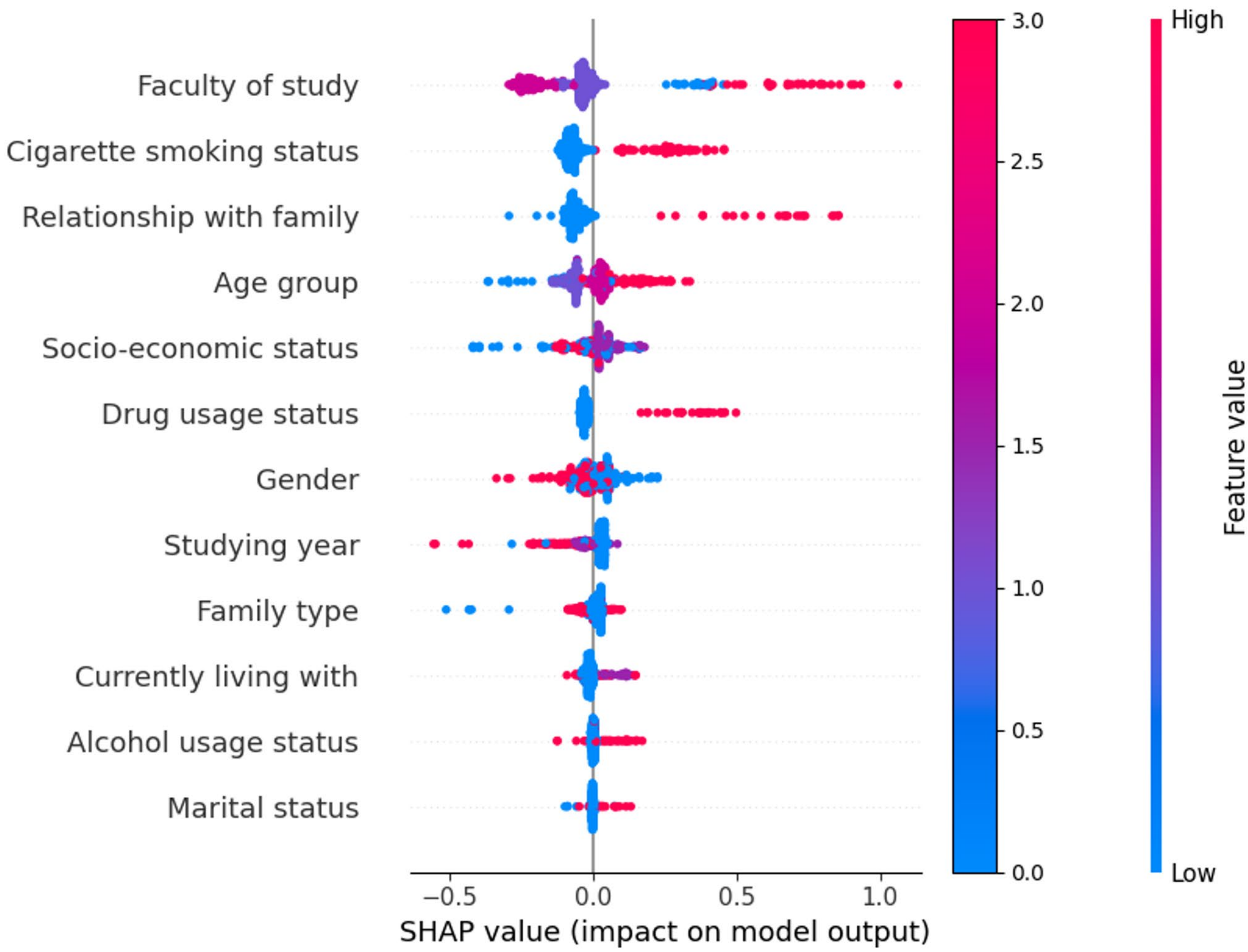

Fig. 2 Features' impact on the model by the CatBoost SHAP value

type had lower scores, suggesting that they have less of an impact on the model's results. This feature importance ranking makes it easier to interpret the model's decision-making process by identifying the most predictive factors.

Optimized model parameters

Table 3 presented the average optimized parameters for the models across the three outcomes, depression, anxiety, and stress. For KNN, the optimal number of neighbors varied between 5.8 for depression, 6.6 for anxiety, and 7 for stress. RF showed consistent optimal parameters with a maximum depth of 10 and 160 estimators for depression and stress, while for anxiety, the number of estimators was slightly reduced to 140. SVM performed best with a regularization parameter (C) and gamma both set to 0.1 for all outcomes. In GBM, the optimal learning rate was 0.05 for depression and anxiety, and 0.06 for stress, with a maximum depth of 3 and 120 to 100 estimators across the models. For XGBoost, the learning rate was set to 0.05 for anxietyand stress, while for depression, it was increased to 0.07, with a fixed max depth of 3 and 3.4. CatBoost optimized its parameters with depth ranging from 4.0 to 5.6, learning rates between 0.06 to 0.08, and 100 iterations for all outcomes. Finally, LR was optimized with a C value of 0.1 for depression and 0.64 for anxiety and 2.26 for stress. These parameters were chosen after evaluating the model performance on various metrics, ensuring the best predictive capabilities for each mental health outcome.

Evaluation of machine learning model performances

Table 4 shows the results of evaluating the effectiveness of various machine learning models for predicting depression, anxiety, and stress statuses based on the average performance metrics across 5-folds. For depression, the SVM model achieved the highest accuracy of 56.93%, precision of 75.60%, and log loss of 0.6847, whereas XGBoost performed the best with an F1-score of 0.5317. In terms of anxiety, SVM showed excellent performance, achieving the highest accuracy (69.48%) and precision (78.81%), while LR also demonstrated competitive

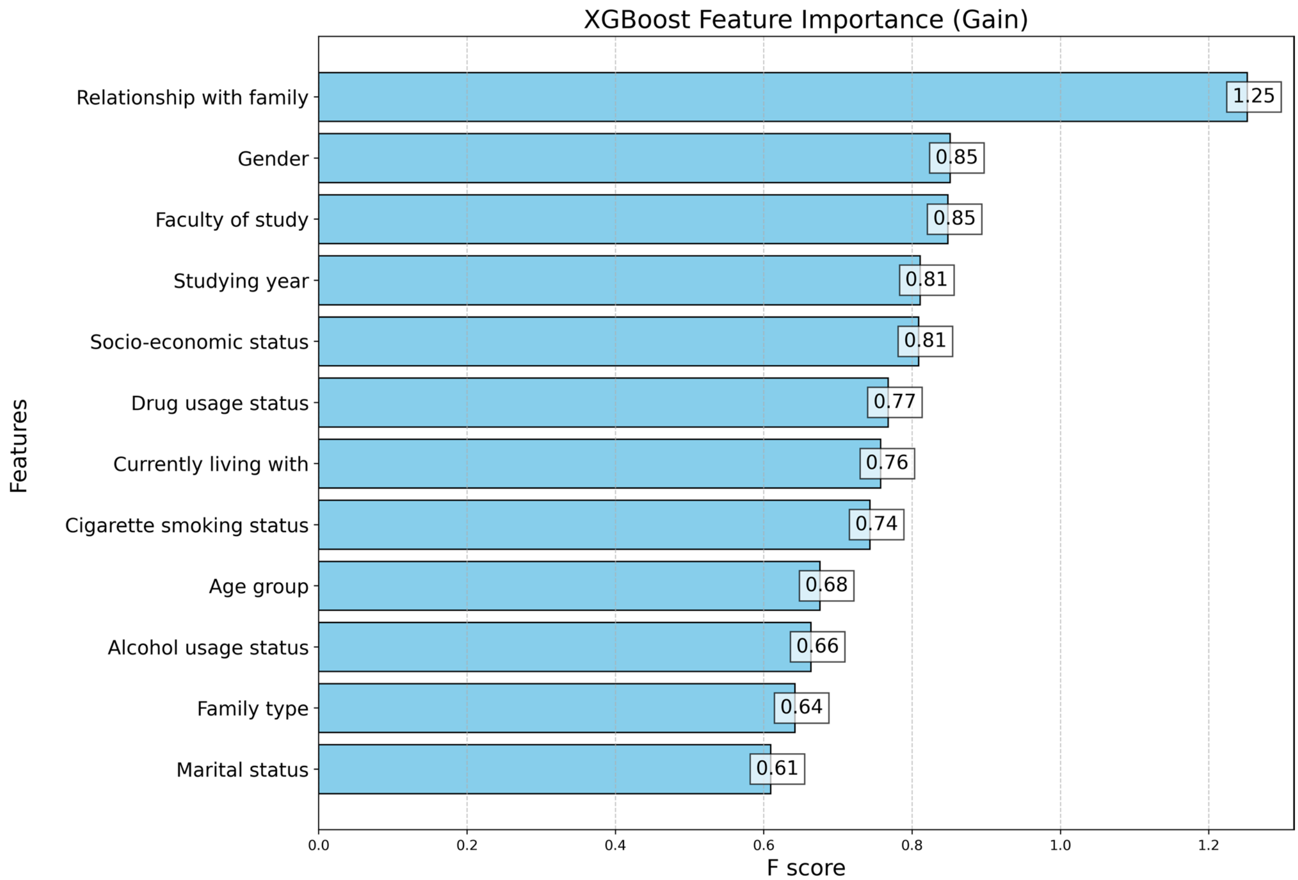

Fig. 3 Features' impact on the model by XGBoost Gain value

Table 3 Average optimized parameters for mental health outucome

Model	Depression	Anxiety	Stress
KNN	n_neighbors = 5.8	n_neighbors = 6.6	n_neighbors = 7
RF	max_depth = 10, n_estimators = 160	max_depth = 10, n_estimators = 140	max_depth = 10, n_estimators = 160
SVM	C = 0.1, gamma = 0.1	C = 0.1, gamma = 0.1	C = 0.1, gamma = 0.1
GBM	learning_rate = 0.05, max_depth = 3, n_estimators = 120	learning_rate = 0.05, max_depth = 3, n_estimators = 100	learning_rate = 0.06, max_depth = 3, n_estimators = 100
XGBoost	learning_rate = 0.07, max_depth = 3, n_estimators = 140	learning_rate = 0.05, max_depth = 3.4, n_estimators = 100	learning_rate = 0.05, max_depth = 3, n_estimators = 100
CatBoost	depth = 4.8, iterations = 100, learning_rate = 0.08	depth = 4.0, iterations = 100, learning_rate = 0.07	depth = 5.6, iterations = 100, learning_rate = 0.06
LR	C = 0.1	C = 0.64	C = 2.26

results, with an accuracy of 69.48% and precision of 78.81%. In terms of stress, CatBoost outperformed the other models with the lowest log loss (0.6284), and it also showed strong performance with an accuracy of 67.06%, a precision of 64.54%, an F1-score of 57.77%, and a low log loss of 0.6284. However, CatBoost exhibited the lowest log loss of 0.6136 for anxiety, indicating superior model calibration compared to the other models. Overall, CatBoost consistently showed the best log loss across all outcomes, while SVM and LR excelled in predicting depression and anxiety, and CatBoost was most effective for predicting stress.

To predict depression, anxiety, and stress status, Fig. 4 shows the ROC curves for several machine learning models. Plotting the trade-off between true positive and false positive rates, each curve shows how well the models perform; AUC values near 0.5 suggest that the models have limited predictive power. Each plot represents the true positive rate (TPR) against the false positive rate (FPR) for different models. The Area Under the Curve (AUC) for each model is annotated in the legend. For depression, RF and GBM achieved the highest (both with AUC = 0.56), followed closely by SVM (AUC = 0.55). LR had the lowest AUC (0.50), indicating poor predictive capability. For

Table 4 Average performance metrics of ML models for predicting mental health outcomes

Model	Depression				Anxiety				Stress			
	Accuracy	Precision	F1 score	Log loss	Accuracy	Precision	F1 score	Log loss	Accuracy	Precision	F1 score	Log loss
KNN	0.4879	0.5335	0.4698	7.3441	0.5675	0.5956	0.5775	4.1179	0.6641	0.6117	0.5750	4.3939
RF	0.5569	0.5386	0.5253	0.6802	0.6901	0.5902	0.5813	0.6170	0.6635	0.6172	0.5910	0.6369
SVM	0.5693	0.7560	0.4136	0.6847	0.6948	0.7881	0.5697	0.6148	0.6676	0.7792	0.5351	0.6325
GBM	0.5645	0.5521	0.5246	0.6815	0.6918	0.6136	0.5805	0.6166	0.6570	0.6033	0.5791	0.6366
XGBoost	0.5687	0.5581	0.5317	0.6870	0.6889	0.5931	0.5711	0.6159	0.6647	0.6175	0.5796	0.6305
CatBoost	0.5598	0.5436	0.5055	0.6798	0.6942	0.7504	0.5715	0.6136	0.6706	0.6454	0.5777	0.6284
LR	0.5593	0.5248	0.4483	0.6831	0.6948	0.7881	0.5697	0.6170	0.6629	0.6026	0.5580	0.6307

anxiety, KNN performed best (AUC = 0.57), followed closely by RF and XGBoost (both with AUC = 0.55), while SVM and LR performed less effectively with AUC values of 0.50. In the case of stress, all models showed a more uniform performance with AUC values ranging from 0.56 to 0.60, with LR achieving the highest AUC (0.60), followed by RF, XGBoost, and CatBoost (each with AUC = 0.59). These results suggest that KNN and RF were more effective in predicting depression, while LR showed the best performance for stress, although the AUCs across all models were relatively close for each mental health outcome. The AUC values for all models and mental health conditions are approximately 0.5, suggesting that there is little discriminatory power in correctly classifying mental health statuses. This could suggest that additional model improvement, better feature selection, or different modeling techniques are required to improve predictive performance.

Discussion

Mental health is a crucial component of overall well-being, encompassing an individual's psychological, emotional, and social state. Despite its importance, mental health conditions are increasingly prevalent worldwide, particularly among young adults such as university students. These conditions pose significant public health challenges, often resulting in impaired academic performance, social isolation, and heightened suicide risks. The university environment can be uniquely stressful, as students often face pressures related to academic performance, future career uncertainty, and adapting to new social networks. This study addresses a gap in the existing literature by comprehensively examining the factors associated with depression, anxiety, and stress among Bangladeshi university students, using an integrated approach that combines traditional statistical methods with advanced machine learning techniques. The study's unique contribution lies in its identification of risk factors, including gender, family relationships, academic pressures, and substance use, and the application of machine learning models (CatBoost, XGBoost, SVM, RF, KNN, LR, and GBM) to predict mental health risks. By integrating both conventional and advanced machine learning approaches, our findings offer a more comprehensive understanding of mental health risk in this population and highlight the value of multidisciplinary research in this area. The findings from this study will inform the development of targeted interventions and policies to improve student well-being in Bangladesh and similar low- and middle-income countries.

The prevalence of mental health conditions among the study participants was notable, with 56.9% experiencing depression, 69.5% anxiety, and 32.2% stress. These rates are generally higher than those reported in previous

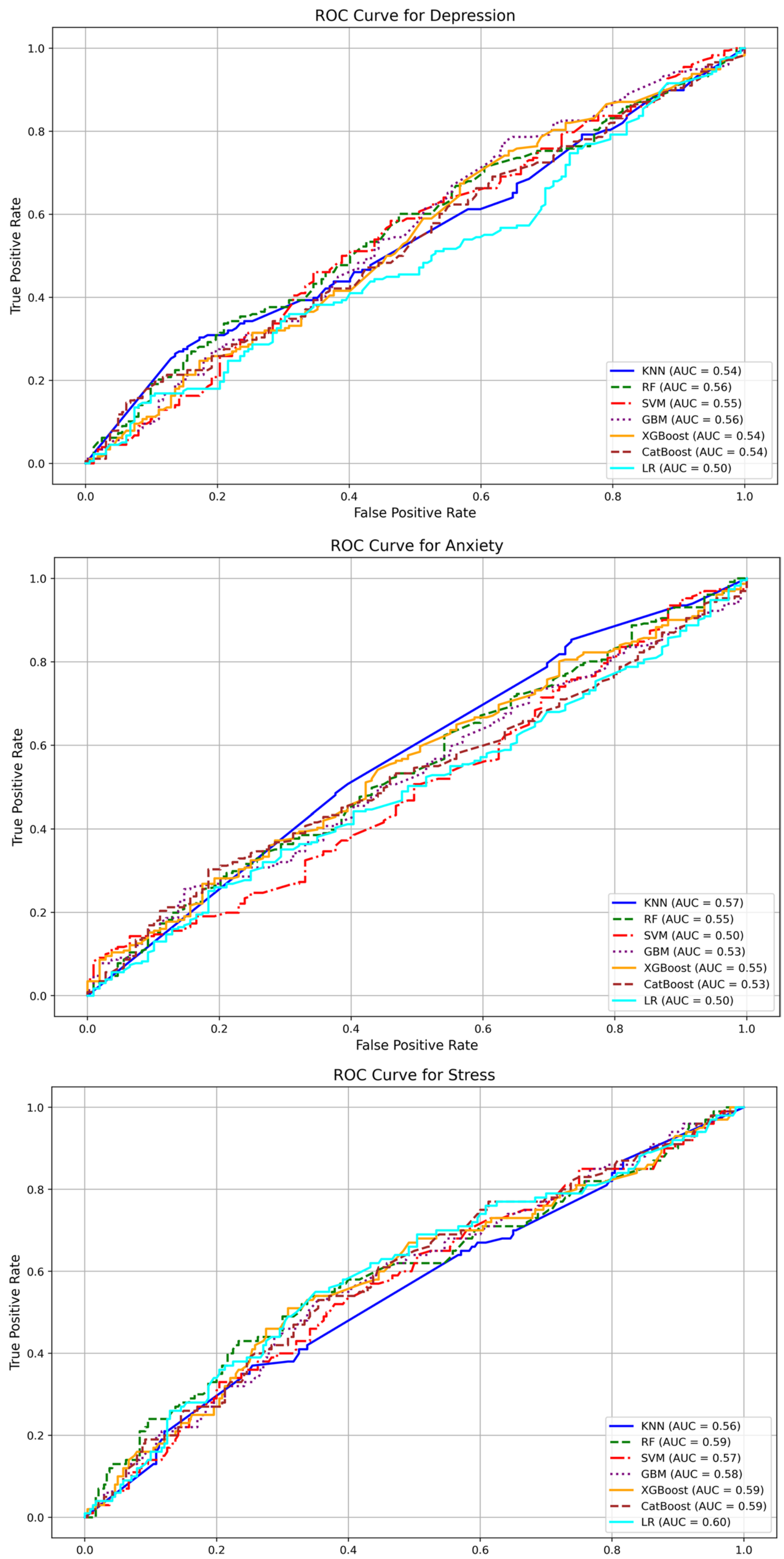

Fig. 4 ROC curve of depression status, anxiety status, and stress status

studies conducted in other countries, such as the USA (depression 22%) [5], China (depression 23.45%) [8], Malaysia (depression 29%) [9], Spain (depression 18.4% vs. anxiety 23.6% vs. stress 34.5%) [10], except in discordant findings in Pakistan (anxiety 88.4% vs. depression 75%) [4]. Among the university students. In the context of Bangladesh, previous studies conducted during the COVID-19 pandemic reported depression, anxiety, and stress prevalence rates ranged from 46.92 to 82.4%, 26.6–96.82%, and 28.5–70.1% [41–45]. These findings align more closely with those of the present study, suggesting a possible pandemic-related impact on the mental health of university students. Moreover, these high prevalence rates might reflect ongoing social and academic pressures, limited mental health resources, and cultural factors that may discourage help-seeking behaviors. Furthermore, a meta-analysis on mental health problems during the COVID-19 pandemic in Bangladesh revealed pooled prevalence rates of 47% for depression, 47% for anxiety, and 44% for stress [27]. These figures not only reinforce the high burden of psychological distress in Bangladeshi populations during and after the pandemic but also contextualize the elevated rates observed in our current study.

The gender-based comparison in this study aligns with previous research [10, 46], indicating that females were more likely to exhibit higher levels of depression and stress than their male counterparts. This finding is consistent with the broader literature reporting women's vulnerability to anxiety and depression [47, 48]. Previous epidemiological studies have also suggested that females are more prone to psychological symptoms than male students [48–50]. Factors contributing to the higher prevalence of mental health conditions in women may include the heavy academic workload [50], academic pressure [51], and limited job opportunities compared to males. Additionally, women may experience greater societal expectations and restrictions, which can contribute to chronic stress and exacerbate underlying vulnerabilities to mental health problems. The social and intellectual challenges encountered during higher education can exacerbate emotional strain, potentially leading to the manifestation of mental health issues [52]. These stressors and societal challenges faced by female students may explain the higher rates of mental health conditions observed in the current study. In contrast, one study reported a higher prevalence of depression among male students than female students [53]. This difference suggests that gender-related mental health risks may be influenced by complex interactions between personal, academic, and cultural factors.

A notable finding of this study pertains to familial relationships and their association with mental health outcomes. Participants who reported having an unfriendly or strained relationship with their family exhibited significantly higher levels of depression, anxiety, and stress. This aligns with prior research, such as a study conducted among Jordanian students, which found that positive family functioning was inversely associated with these mental health conditions [54]. These findings are further supported by other literature highlighting the critical role of family support in fostering psychological well-being [46]. In the context of Bangladesh, where family plays a central role in students' emotional and social lives, strained relationships may contribute to a sense of isolation and a lack of coping resources. This study expands upon this body of work by emphasizing that poor familial relationships are a contributing factor to mental health problems. Consequently, interventions aimed at improving family dynamics may serve as an effective strategy for mitigating these issues. This association is particularly salient within the context of Bangladeshi culture, where family plays a central role in the emotional and social lives of individuals, thereby significantly influencing mental health outcomes. A lack of emotional support from family may foster a sense of isolation, further exacerbating psychological distress among young people. This is consistent with other studies that have shown limited familial and community engagement to be predictive of adverse mental health outcomes [55]. Universities may benefit from engaging families through mental health education and outreach programs, fostering supportive networks beyond the campus.

The present study found that depression and anxiety, but not stress, were significantly associated with students' academic disciplines. Students from commerce faculties reported the highest prevalence of depression (73.9%) and anxiety (69.7%) compared to those in other faculties, which is in line with a previous study reporting similar patterns [12]. Additionally, another study suggests that concerns such as perceived career uncertainty and feelings of personal inefficacy are commonly expressed by students in these disciplines [56], which may be relevant in interpreting these findings. The higher rates of distress in commerce students may reflect worries about employment prospects, economic instability, and intense competition in the job market. In terms of academic year, a significant association was observed with stress, but not with depression or anxiety. Third-year students reported the highest stress levels (41.4%), while first-year students reported the lowest (32.3%). This observation is consistent with earlier research indicating that academic demands and workload may increase throughout university education, possibly contributing to elevated stress levels in later years [57, 58]. Support programs for students in advanced academic years, such as mentorship and academic counseling, may help to alleviate these stressors and reduce risk. However, no significant association was observed between mental health outcomes and socioeconomic status in this study. This contrasts with findings from previous studies that often report lower and higher socioeconomic status as a risk factor for mental conditions [59]. The relatively homogenous background of the current sample, mainly comprising university students, may have limited the variability needed to detect such associations. It is also possible that strong peer networks or campus resources may buffer the impact of socioeconomic differences within this group.

The overall findings of this study indicated significant associations between health and behavioral risk factors and mental health outcomes. Specifically, substance use behaviors, including cigarette smoking, illicit drug use, and alcohol consumption, were significantly associated with symptoms of depression and anxiety. These results are in line with a substantial body of literature demonstrating links between substance use and mental health issues. For example, prior research has reported that heavy use of alcohol and drugs is associated with increased risks of depression, anxiety, and cognitive impairment among adolescents and young adults [60]. Psychologically, some students may use substances as a way of coping with academic or emotional stress, but this may in turn aggravate mental health issues, creating a vicious cycle. The current findings contribute to this growing body of evidence by documenting similar patterns within a Bangladeshi university student population. The observed comorbidity between mental health conditions and substance use may be interpreted in light of both biological and psychosocial frameworks. Some studies have suggested that substance use may influence brain chemistry in ways that exacerbate symptoms of anxiety and depression, while others point to the potential for individuals with pre-existing mental health concerns to engage in substance use as a form of self-medication [60]. In the context of cigarette smoking, the relationship appears to be bidirectional, with mental health challenges potentially leading to smoking as a coping strategy, and smoking itself possibly contributing to worsening psychological symptoms [61–63]. Prevention and intervention programs should consider this complex interplay and incorporate both mental health promotion and substance use prevention as integrated strategies for university students. These findings highlight the need for integrated approaches that address both substance use and mental health among young people. Including smoking cessation and substance use prevention as components of mental health interventions could be particularly relevant for university students in Bangladesh.

The application of machine learning techniques in the current study offered an additional layer of analysis to identify patterns associated with depression, anxiety, and stress among university students. SVM demonstrated the best performance for depression, achieving the highest accuracy (56.93%) and precision (75.60%), while also exhibiting a relatively low log loss (0.6847). In contrast, KNN performed the worst in depression, with the lowest accuracy (48.79%) and precision (53.35%). However, SVM showed strong results for anxiety, with an impressive accuracy (69.48%) and precision (78.81%), making it one of the top performers in this category. Similarly, LR performed decently in both anxiety, achieving relatively high accuracy and log loss scores across all models. For stress, CatBoost was particularly noteworthy, achieving the highest accuracy (67.06%) and precision (64.54%), while RF and XGBoost showed promising results in stress prediction, particularly in terms of log loss (0.6369 and 0.6305, respectively). Overall, SVM, RF, and CatBoost emerged as the top contenders in predicting depression, anxiety, and stress. These findings underscore the variability in model performance across different mental health outcomes, with no single model consistently outperforming others across all metrics. Its effectiveness may be attributed to its algorithmic strengths in handling categorical variables and reducing overfitting, as supported by existing literature [64]. These results indicate that, while machine learning can add value in identifying students at risk for poor mental health, predictive accuracy is still limited, possibly due to the complexity of mental health determinants and the available data. Despite these observations, receiver operating characteristic analysis revealed that the discriminative ability of all models was limited, with area under the curve values not exceeding 0.54 for any of the mental health outcomes. These results suggest that the predictive capacity of the models remains modest, potentially due to the complex and multifactorial nature of mental health conditions, which may not be fully captured by the available feature set. Future studies could benefit from incorporating additional features, such as digital behavioral markers or longitudinal data, to improve model performance. Nevertheless, the k-nearest neighbors algorithm yielded the highest F1 scores across outcomes, indicating a relatively balanced trade-off between precision and recall in classifying individuals with varying mental health symptomatology.

Feature importance analyses offered valuable insights into the most influential predictors. For instance, CatBoost identified cigarette smoking status as a key predictor across outcomes, whereas XGBoost highlighted relationship quality with family (feature importance = 1.30) as a significant correlate. These findings are consistent with earlier literature emphasizing the role of behavioral and psychosocial factors in mental health risk among youth. The integration of SHAP (SHapley Additive exPlanations) values further enhanced model interpretability, revealing feature-specific contributions to prediction and reinforcing the context-specific relevance of certain variables within the Bangladeshi student population. When benchmarked against international studies, such as the work by Zhai et al. [65], which achieved notably higher AUC values (0.74–0.77) by incorporating contextual socioeconomic and environmental variables in a large U.S. student cohort, the current models demonstrated comparatively limited performance. This discrepancy may highlight the need for a broader set of features, including dynamic or time-sensitive variables such as financial insecurity, academic stress trajectories, or campus support engagement, to better capture the complex determinants of mental health outcomes. Furthermore, differences in data collection methods, sample composition, and cultural context may contribute to the observed differences in model performance, emphasizing the need for context-specific model development and validation.

In recent years, several ML studies in Bangladesh have focused on predicting mental health outcomes among university students, particularly concerning depression, anxiety, and stress. For example, a study utilized seven machine learning models, such as Random Forest, Support Vector Machine, and Artificial Neural Networks, to analyze data from 750 students and found the RF as the best-performing model achieved an 87% accuracy, 78% precision and 86% f1-score rate in identifying depressive symptoms [66]. Another study implemented machine learning algorithms, including Decision Tree, RF, and SVM, to predict and classify perceived stress and that study identified significant predictors, including pulse rate, diastolic blood pressure, sleep status, and smoking status, where the RF model performs better with 0.8972 accuracy, 0.9241 precision and 0.8715 AUC-ROC [23, 67]. Furthermore, research that concentrated on the prediction of anxiety and depression employed eight machine learning algorithms. RF, Decision Tree, and AdaBoost achieved 100% accuracy and precision, while SVM exhibited the quickest performance time with 0.007 s [67]. The potential for early detection and intervention strategies is underscored by these studies, which emphasize the increasing implementation of ML techniques in mental health research within the Bangladeshi university context. Despite these technological advances, ensuring ethical use, transparency, and user trust in ML-based mental health tools remains a priority for research and practice.

This study's findings have several important implications for university policy and mental health practice in Bangladesh and similar contexts. The high prevalence of depression, anxiety, and stress among university students—together with the identification of specific risk factors such as unfriendly family relationships, female gender, advanced academic year, and substance use—signals an urgent need for comprehensive and systematic mental health support at the institutional level. Universities should develop and implement evidence-based mental health policies that prioritize routine mental health screening, particularly targeting students with identified risk factors. Such screening could be integrated into regular student health services, orientation programs, or academic advising, enabling early identification and timely intervention for those at risk. The expansion of accessible, confidential, and culturally appropriate on-campus counseling services is also critical, with adequate staffing and training to address the diverse needs of the student population. Establishing structured peer support programs and student-led mental health clubs can further foster a supportive community, reduce stigma, and encourage help-seeking behaviors. Policies should promote collaboration between faculty, administration, and health professionals to ensure that mental health support is embedded across all aspects of campus life. In addition, universities should integrate mental health education into their curricula and co-curricular activities to enhance mental health literacy, resilience, and self-care skills among students. Awareness campaigns and anti-stigma initiatives can create a more open environment for discussing mental health concerns. At a broader policy level, the Ministry of Education and relevant governmental bodies should recognize student mental health as a priority, allocate resources accordingly, and develop national guidelines for university mental health services. Given the demonstrated utility of machine learning models in identifying students at risk for mental health problems, universities and policymakers should consider leveraging data-driven approaches—such as electronic risk assessments and early warning systems—to supplement traditional support mechanisms. Integrating these digital tools, while ensuring privacy and ethical use of student data, could enable more proactive and individualized support. Ultimately, a multi-level approach that combines institutional policy reform, resource allocation, technological innovation, and community engagement is needed to address the mental health crisis among university students in Bangladesh. These strategies can serve as a model for other low- and middle-income countries facing similar challenges in student mental health.

This study highlights the effectiveness of tree-based and boosting machine learning models in predicting mental health outcomes, offering advantages over traditional regression methods by capturing nonlinear relationships and interaction effects among predictors. Such capabilities enhance both the predictive accuracy and interpretability of complex behavioral health data. With a sample size of 1,697 students and balanced outcome classes, the models performed reliably within this context. However, the generalizability of these findings may be limited in studies with smaller or imbalanced datasets, where classification performance could decline. Additionally, as a cross-sectional design was employed, causal inferences cannot be drawn, underscoring the need for future longitudinal studies. The exclusive reliance on self-reported data also raises the potential for response bias; thus, future research should integrate objective measures of mental health. While the machine learning approaches used allowed for the exploration of intricate variable relationships, model interpretability remains a challenge and requires further refinement. Additionally, this study was limited to two public universities (Jahangirnagar University and Patuakhali Science and Technology University), and our sample did not include private universities. Thus, the findings may not be generalizable to students at private institutions, where mental health outcomes and influencing factors may differ [68]. Furthermore, Use of convenience sampling despite stratified quotas means participants were not selected with a known probability, so estimates may be affected by selection and volunteer bias. Coverage was limited to Jahangirnagar University, Dhaka, and Patuakhali Science and Technology University, Patuakhali, and specific recruitment times, potentially under-representing students who were absent, off-campus, or less engaged with classes. Non-response and self-selection may distort associations, and findings should not be generalized to all university students in Bangladesh without caution. Future studies should employ probability-based designs (e.g., multistage cluster sampling from a national or registrar sampling frame with stratification by university type, faculty, and academic year), implement non-response follow-up, and apply post-stratification weights to improve external validity. Future research should include diverse university types to provide a more comprehensive understanding of student mental health across Bangladesh. These findings emphasize the importance of developing targeted interventions addressing familial relationships, substance use, and academic stress to improve youth mental health in Bangladesh.

Conclusions

This study depicts important evidence regarding the prevalence and correlates of mental health issues among Bangladeshi youth. The findings reveal that poor family relationships, gender disparities, substance use, and academic pressure have been identified as major contributory risk factors for mental health problems in youth. Teens in families that struggle to connect with one another could experience a surge of anxiety, depression, and stress conditions due to the extra scholastic pressure on them or because they have resorted to substance consumption as some kind of outlet. The findings suggest the necessity for culture-specific mental health interventions that target and modulate these identified risk factors. Policymakers can use these findings to create strategies and initiatives that target strengthening family relationships and offering gender-based mental health support while developing academic stress management mechanisms. Collaboration among policymakers, educators, and individuals in the mental health professions is necessary to establish school-based awareness programs to yield maximum benefits on youth depression. Longer-term research should explore the effectiveness of family-focused interventions, substance use prevention programs, and academic stress reduction strategies on mental health outcomes for youth. Further, investigating the influence of socio-economic status, peer connections, and accessibility to mental health services will enrich our understanding of determinants for youth in Bangladesh.

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Author contributions
MEH, MA, and SMRH were involved in writing the original draft, reviewing and editing the manuscript, developing the methodology, conducting the investigation, administering the project, and performing formal analysis. MM, JA, MMK, MMA, PAA, MM, FAM, and MAM contributed to the methodology, writing – review and editing, supervision, provision of resources, and validation. FAM and MAM also contributed to the conceptualization, investigation, project administration, data curation, writing – original draft, and writing – review and editing. All authors read and approved the final manuscript.

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Data availability
The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Human Ethics and Consent to Participate
The study meticulously followed ethical guidelines outlined in the Helsinki Declaration of 2013 to safeguard the welfare of human participants. Ethical clearance was obtained from the review board at Patuakhali Science and Technology University [Reference: PSTU/ICE/2023/83], before the study's initiation. Before their involvement, all participants received comprehensive information about the study's objectives, procedures, and potential risks and benefits.Informed written consent was obtained from each participant before survey administration, emphasizing their voluntary participation and right to withdraw from the study at any point.

Consent for publication
This study participation required informed written consent from the participants.

Competing interests
The authors declare no competing interests.

Clinical trial number
Not applicable.

Author details
[1]CHINTA Research Bangladesh, Savar 1342, Dhaka, Bangladesh
[2]School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China
[3]Department of Ayurvedic Medicine, Institute of Teaching and Research in Ayurveda, Jamnagar, Gujarat, India
[4]Department of Psychology, University of Dhaka, Dhaka 1000, Bangladesh
[5]Department of Psychiatry, Mbarara University of Science and Technology, Mbarara, Uganda
[6]Department of Psychiatry, King Ceasor University, Kampala, Uganda
[7]Department of Psychiatry, Uganda Christian University, Kampala, Uganda
[8]Department of Psychiatry and Behavioural Neurosciences, McMaster University, Hamilton, Canada
[9]Department of Psychology, College of Education and Human Development, Princess Nourah Bint Abdulrahman University, Riyadh, Saudi Arabia
[10]Institute of Psychology, Faculty of Social Sciences, University of Gdansk, Gdansk 80-309, Poland
[11]Department of Public Health, University of South Asia, Dhaka, Bangladesh
[12]Department of Public Health & Informatics, Jahangirnagar University, Savar 1342, Dhaka, Bangladesh

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